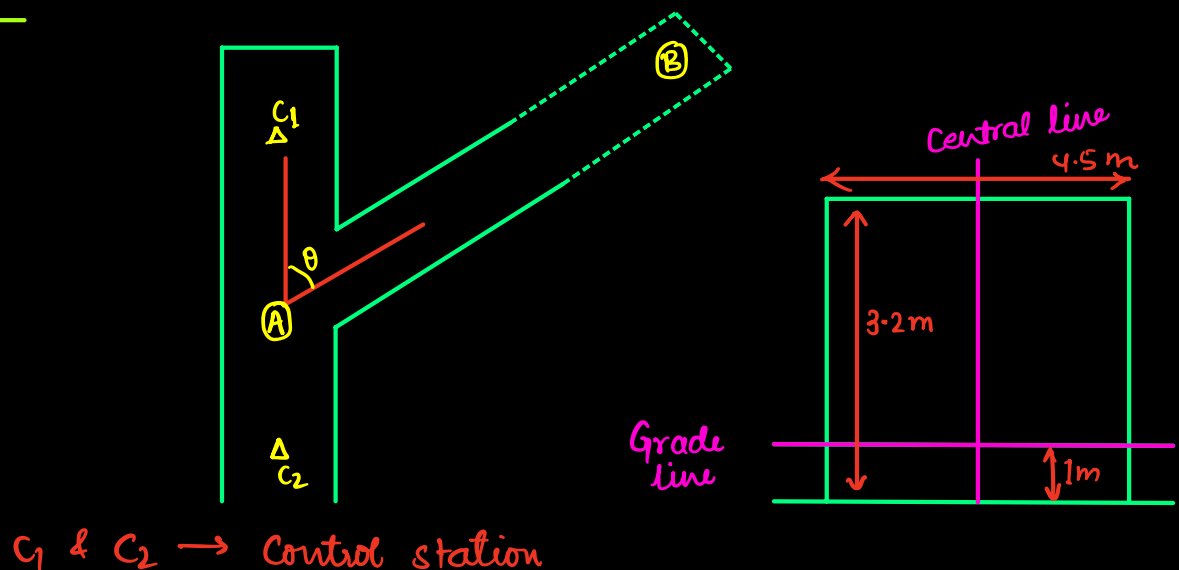


A drift of 100 m length and opening size 4.5m x 3.2 m is to be driven from Point "A" to "B" in an underground metal mine using drilling and blasting method of excavation. Drift is to be developed using trackless mining system with single boom drill jumbo of 3.2 m drill rod, LHD and LPDT combination of 5 tonne and 15 tonne capacity, respectively, etc. Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed:

- i. Steps involved in the development of the Drift
- ii. Manpower distribution plan activity wise/shift-wise
- iii. Days required to complete the development of the drift
- iv. Equipment/Machineries deployed/needed
- v. Drilling and blasting pattern
- vi. Drilling meterage per round of blasting
- vii. Explosive consumption per round of blasting
- viii. Tonnage produced per round of blasting
- ix. Powder factor obtained
- x. Ventilation circuit for ventilating the face

Solution —————
(i)



Steps involved :—

- ① Survey of the proposed location as per the plan and marking central line and grade line (for direction & gradient)
- ② Drilling ———
Drilling with drill jumbos or Jack hammer in the proposed direction and gradient.
Burn cut for hard rock

Wedge cut for soft rock

Length of hole is 3.2m (drill jumbo)

" " 8ft/2m (Jack hammer)

③ Charging & blasting —

charging with emulsion explosive in the form of cartridge, Electric delay detonators are used.

Blasting with the help of exploder from a safe distance atleast one right angle from the face direction.

Blasting is done at the end of the shift. During blasting auxiliary fan should remain off.

④ Fume clearance —

Fume clearance is done with the help of auxiliary forcing fan / combination of forcing & exhaust fan.

⑤ Water spraying & loose dressing —

Water spraying & loose dressing is done manually. Water spraying suppress the dust and loose dressing operation helps in separating loose rock generated after blasting in the back & sides of the face.

⑥ Loading & Hauling of blasted material —

Loading & hauling operation is carried out using LHD & LPDT combination in trackless mining system whereas Tub / Grizzly Car & Rockerz shovel / Loader (pneumatically operated) used in track bound mining system. Loaded material is dumped near by V/G void (if available)

otherwise hoisted to surface and dumped in waste dump yard.

⑦ Face preparation) Ready for next round of drilling

⑧ Support of back & side (if required) ———

After three rounds of blasting about 9m progress, support for back as per support plan (SSR) framed by Mine with the approval of DGMS. Generally roof bolt of 2m is used to support the back in the grid of 1.2m & 1.2m. (Cement + sand + water) is used as a grouting material. Full column grouting is done.

⑨ Repeation of the process ———

(Drilling → Charging & Blasting → Face clearance
→ water spraying & loose dressing → loading & hauling of blasted rock → Face Ready → Support)

★ Development Survey ———

Fixing Centre peg & grade peg & extending the central line & grade line.

In order to maintain direction and gradient of the excavation/ Development type, after face advancement around 20-25m (In case of drill jumbo face) and around 10-15m (In case of Jack hammer face), centre pegs & grade pegs are fixed by the surveyors in the back & sides of the face resp. A set of new centre pegs (2 Nos.) and grade pegs (2+2=4 Nos.)

is established after 20~25m face advancement.

Measuring size of the opening (HxW) —

In order to assess the size of the opening as per the actual size, Development survey is done. It helps in monitoring the overbreakage or underbreakage (both are undesired).

* Drainage —

Drainage is to be provided to drain the water from the face (For decline & Ramp down, face pump is required during drilling & charging). For drift / drive, it is developed in up-gradient (1 in 200) and drain is required. For ramp up & ramp down, no need of drain. For decline, drainage is made at one side of the floor.

* Extension of auxiliary services —

Water pipeline, compressed air pipe line, electrical cable, gate end box, manhole ventilation duct are extended at regular interval as face advances.

(ii) Manpower distribution plan activity wise / shift wise —

1 shift

① Survey & Marking → 1 sup + 2h + 1MM + 1FM
(A) (7AM-3PM)

3 shifts

② Drilling → 1 DJO + 1h + 1MM + 1FM
(B) (3PM-11PM)

{
②+③+
⑤+⑥

- 3 shifts per round of blasting
- ③ {8AM - 4PM} Charging & Blasting → 1B + 2h + 1MM
 (B) (3PM - 11PM)
 - X ④ Fume clearance → No manpower required
 - ⑤ Water spraying & Loose chocking → 1TM + 1ATM + 1MM + 1FM
 (C) (11PM - 7AM)
 - ⑥ Loading & Hauling → 1TM + 1ATM + 1MM + 1FM
 (C) (11PM - 7AM)
 - ⑦ Face preparation → 1THO + 4h + 1MM + 1FM + 1B
 (B) (3PM - 11PM)
 - ⑧ Support → 1TM + 2h + 2MM + 1FM
 (C) (11PM - 7AM)

Development Survey —

- 1 shift for 20m
 ↓
 5 shifts for 100m
- ⑨ (i) Fixing pegs & extending C/L & G/L } → 1sur + 2AS + 1MM + 1FM
 (ii) Measuring opening size
 (A) (7AM - 3PM)
 - ⑩ Drainage → 1RBO + 1h
 (A) (7AM - 3PM)

- ⑪ Extension of Auxiliary services —

- 3 shifts
- (i) Water pipe line & compressed air pipe line extension $\longrightarrow 1PF + 1h$
 (A) (7AM - 3PM)
 - (ii) Electrical Cable extension & GEB extension $\longrightarrow 1Elec. + 1h + 1MM$
 (A) (7AM - 3PM)

- 1 shift for 20m
- (iii) Manhole $\longrightarrow$ Jack hammer crew (1THO + 2h) + 1MM + 1PM
 (A) (7AM - 3PM)

- 11 shifts
- (iv) Loading Bay $\longrightarrow 1PF + 2h$
 (A) (7AM - 3PM)

(v) Ventilation duct extension

Total shift needed for 100m

$$= \frac{100 \times 3}{3.2 \times 0.9} + \frac{100}{3.2 \times 0.9} \times \frac{2}{3} + 1 + 3 + 11 + 5 + 5$$

$$= 104 + 23 + 25$$

$$= 152 \text{ shifts}$$

Total Shifts after considering B/d time + unforeseen situation

$$= 152 + 152 \times \frac{5}{100}$$

$$\approx 160 \text{ shifts}$$

(iii) No. of Days required ———

$$\text{In days ; } \frac{160}{3} = 53 \text{ days}$$

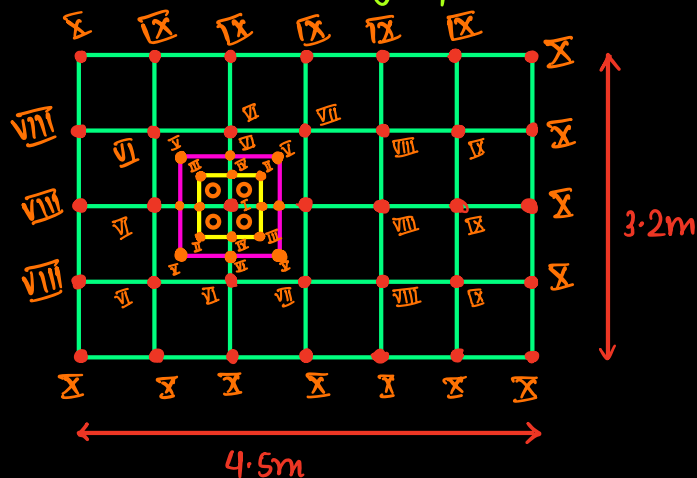
$$\text{In month ; } \frac{53}{25} = 2.12 \text{ months}$$

(iv) Equipments / machineries required ———

- ① Drill jumbo
- ② LHD/LPDT
- ③ Jack Hammer
- ④ Grouting Gun
- ⑤ Stage / service vehicle
- ⑥ Grade end box (GEB)
- ⑦ Pipeline (Air + water)
- ⑧ Fan
- ⑨ Duct pipe
- ⑩ lighting
- ⑪ Power cable
- ⑫ Face pump (for ramp down, decline)

(v.) Drilling & Blasting pattern ———

Burcut drilling pattern



47+4R

$$\begin{aligned}
 \text{(vi) Total metrage drilled} &= 51 \times 3.2 + 8 \times 3.2 \\
 &= 51 + 8R \\
 &\approx 188 \text{ m}
 \end{aligned}$$

45mm → diameter
65mm ~ 89mm

$$\begin{aligned}
 \text{(vii) Explosive consumption per round} &= 47 \times 3 \\
 &= 141 \text{ kg}
 \end{aligned}$$

$$\begin{aligned}
 \text{(viii) Tonnage per round of blasting} &= 4.5 \times 3.2 \times 3.2 \times 0.9 \times 3 \\
 &= 120 \text{ tonnes}
 \end{aligned}$$

$$\text{(ix) Powder factor} = \frac{120}{141} \text{ tonnes/kg} = 0.85 \text{ tonnes/kg.}$$

b) Characterize the primary and secondary development of underground metal mine based on the Inclination, Location, shape, size, and purposes.

Type of	Development	Inclination	Location	Shape	Size	Purposes
1) Primary	crosscuts	Horizontal	At different location	Rectangular D-shape Arch	As per req. (4.5x3.2)	To connect different drivages/roadways
	Orepass	Vertical/Near vertical	Near V/S	Rectangular circular square		To transfer ore from different levels to ore bin.
	Waste pass	—	—	—	—	—
	Crusher chamber	vertical	below ore bin	Rectangular	As per req.	To crush the ore before hoisting below ground.
	Slump (Main)	Inclined	Near shaft inset, shaft bottom	Rectangular	As per req.	Temporary holding of U/G water
	sub-station	Horizontal	Near shaft inset/In the vicinity of shaft inset	Rectangular	As per req.	To step down the power and supply at to operate different machines.
	Garage	Horizontal	In the vicinity of shaft inset	Rectangular	As per req.	for maintenance of U/G equipment/diesel operated
	End/Raise/Ventilation	Sleep Incline/Vertical/Near vertical	F/W side, sides of the block, as per the req.	Rectangular/square 2x2 m 2.5x2.5m 3.0x2.5m		

Primary Development	Inclination	Location	Shape	Size	Purposes
Orbiter	Vertical	Above crusher chamber	Rectangular	As per req.	To hold ore temporarily
Shaft Inset	Horizontal	At different levels in v/s	Are/ Rectangular	As per req.	To land over a particular level.
Finger Raise	Inclined	At different levels	square/ Rectangular	As per req.	To connect ore pass to levels

Secondary Development	Inclination	Location	Shape	Size	Purposes
Ramps (Up/down)	7° - 9°	F/W side of a stopping block	Rectangular/ D-shape	As per req. (4.5 x 5.2)	For entering of track less eqpt. into the stope. For fresh air to connect.
Access X cut	Horizontal	F/W side, at the centre of the stopping block	- do -	- do -	To connect Drift and ASD
Trough X-cut	Horizontal	F/W side / H/W side at different points	- do -	3m x 3m	To connect F/W Drive to trough drive
Stope Raise	Inclined (> Angle of repose ≥ 40°)	F/W side of ^{stopping} block and at the ends of the block	Square 2m x 2m 2.5m x 2.5m	2m x 2m 2.5m x 2.5m	To ventilate the block as return ways To help in snowing/door filling

Secondary Development	Inclination	Location	Shape	Size	Purpose
ASD	Horizontal	Within the orebody block along the strike, F/W side	Rectangular	As per req. (4.2 x 3.2m)	→ Exploratory mining. → To delineate the orebody
Drill Drive		Within the orebody along the strike	"	"	To facilitate the production drilling from drill level
Trough Drive		—	"	"	" from extraction level
OTR	sub-Vertical / vertical	F/W side of the block	Square (2 x 2m)	"	To transfer ore from
Slot Raise	"	Within the block edge of the block / H/W side	square / rectangular	2.5 x 2.5 / 2 x 2	Stope to lower level free face for blasting / ventilation.

MNC-209

24-01-2024

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a) Briefly Discuss on the following:

- i. Factors influencing development requirements
- ii. Different Purposes served by the mine development
- iii. Excavation Methods and its advantages & Disadvantages

(i) Factors influencing development requirements ———

- orebody type
- Orebody depth
- Mining method selected
- Economics
- Orebody extent

(ii) Mine development serves no. of purposes ———

- Accessing the mineral deposit
- Provision of accurate geological information to aid the production process
- Providing accurate information on other technical needs such as geotechnical characteristics of the rock, water inflow rate, ventilation pollutants inflow & so on.

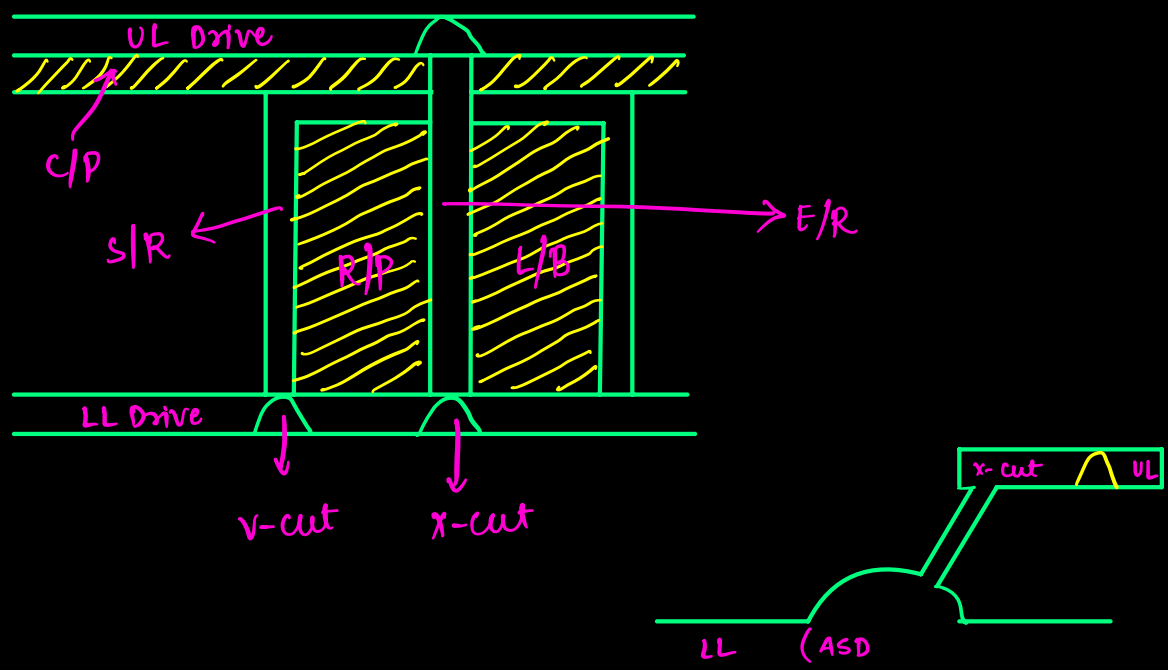
(iii) Excavation Methods ———

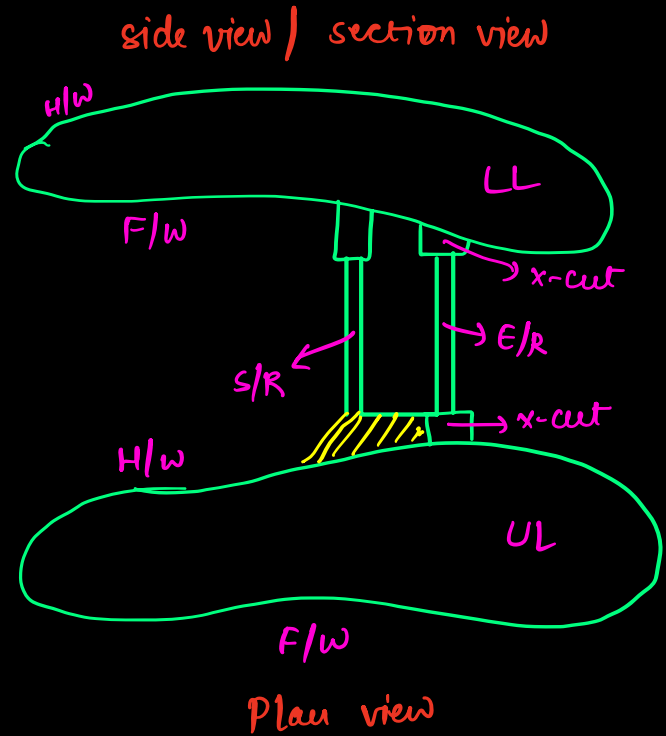
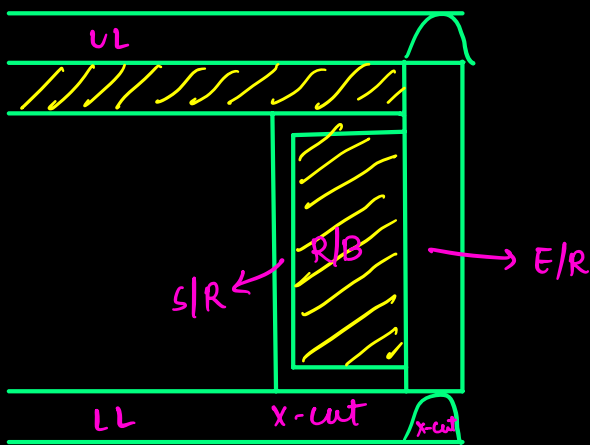
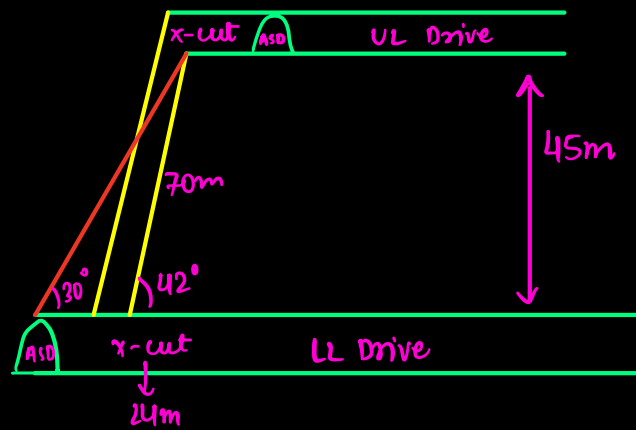
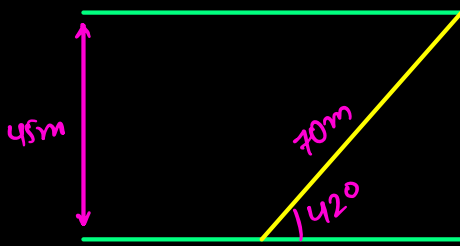
Q.} An end raise/ventilation raise of 70 m length and opening size of 2m x 2m is to be developed in between two levels using Alimak Raise Climber (ARC). Assume your own suitable conditions wherever needed. Briefly explain the following with neat sketches (plan, sections, and figures) wherever needed:

- (i) Steps involved in the raising process using ARC.
- (ii) Manpower distribution plan manwise / shiftwise.
- (iii) Days required to complete raise.
- (iv) Equipments/ Machineries required.
- (v) Drilling & blasting pattern
- (vi) Drilling metrage per round of blasting
- (vii) Tonnage produced per round of blasting
- (ix) Powder factor obtained
- (x) Ventilation circuit for ventilating the face.

Ans: -

For 70 m Raise development using ARC in given condition —





(ii) Steps involved are :—

* Angle guide fixing, service guide fixing, cage fixing, motor fixing, hose reel setup, connection of hose reel with motor etc., platform fixing & canopy fixing.

- ① Preparatory work : water + compressed air pipelines + GEB + lightning + Auxiliary fan + illumination + ventilation duct
- ② Survey & marking (CL/GL) for x-cut (if required).
- ③ Development of x-cut upto the desired length

- ④ Heightening of back of x-cut upto 4~4.5m to accomodate ARC to facilitate mucking operation.
- ⑤ Survey & marking (CL+GL) for proposed raise
- ⑥ 2~3 rounds of blasting in proposed direction and gradient.

⑦ Shifting & installation of ARC —

- Fixing angle guide
- Service guide fixing
- Cage fixing
- Compressed air operated motor fixing
- Hose reel
- Connuction of hose reel with motor
- Platform fixing
- Canopy fixing

⑧ Face marking + face drilling + Charging & Blasting + Fume clearance + water spraying & loose dressing + Face preparation + Repeation of above cycles for 2~3 rounds of blasting

⑨ Service guide rail extension of 2m + face drilling (incomplete)
→ Service guide is fixed with the help of anchor bolt of 2 feet

⑩ Face drilling is completed in next shift.

⑪ Repeation of cycle including service guide rail extension

⑫ Mucking after 3-4 rounds of blasting

⑬ Development survey for C/L and G/L extension along with measurement of opening size after 15-20m face advancement.

⑭ Uninstallation of ARC

⑮ Fitting of wooden ladder in the floor to make it operational.

Manpower distribution plan: —

<u>Activity</u>	Shift			Remarks
	(A)	(B)	(C)	
① Preparatory work - water pipeline + compressed air pipe line + Ventilation duct + Auxiliary fan installation + GEB + lightning, etc.	1 PF + 2h 1 EL + 2h 1 MM +	2-3 shift		
② Survey & marking (C/L & G/L) for x-cut	1 sur + 2h + 1 TM + 1 MM			

③ x-cut development (24m)
+ back/roof heightening
upto 4~4.5m to accomodate
ARC and facilitate mucking
operation

④ Survey & marking 1 sur + 2h +
(CL & GL) for proposed 1 TM + 1 MM
raise

⑤ 2-3 rounds of blasting
in proposed direction
& gradient (4-5 mtr.
progress)

⑥ Shifting & Installation 1 Me + 2h +
of ARC 1 JHO + 2h +
1 MM + 1 PF +
2h + 1 TM } 8-9
shift

⑦ Face drilling + charging 1 JHO + 2h +
& Blasting + FC 1 B + 1h +
1 MM + 1 TM

⑧ WS & LD + FD + 1 TM + 1 MM +
FD (incomplete) 1 JHO + 2h

9) Face drilling + C&B
(complete) 1 JH0 + 2h +
1B + 1h + 1MM
+ 1TM

10) Service Guide rail
exten. + Face drilling
(Incomplete) 1 JH0 + 2h +
1 TM + 1MM

11) Face drilling + C&B +
(complete) FC 1 JH0 + 2h +
1B + 1h +
1TM + 1MM
(B)

12) WS & LD + FP + FD 1 TM + 1MM +
(Incomplete) 1 JH0 + 2h

13) Face drilling + C&B
(complete) + FC 1 JH0 + 2h +
1B + 1h +
1 TM + 1MM
(B)

14) WS & LD + FP + Service
Guide Rail extension +
Face drilling
(Incomplete) 1 TM + 1MM
+ 1 JH0 +
2h
(B)

15) Mucking after 3-4
rounds of blasting

⑩ Development Survey
(CL & GL exten. + opening 1sur + 2h
size meas.) after
15-20m face progress

⑪ Uninstallation of ARC

⑫
1 Mech + 2h +
1 TM + 1 MM } 15 shifts

⑬ Fitting of wooden ladder

1 TM + 2h +
1 MM } 15 shifts

Total days required —

Activity	Shift	Activity	Shift
1.	2-3 shift	7, 8, 9, 10 →	4 blasts in 7 shifts
2	1 shift	11, 12, 13	per day $\frac{4+6}{2} = 5$
3	12 shift	15 (NR)	(considering 2 shift working)
4	1 shift	16	— 1
5	3+3 shift	17	— 15
6	8-9 shift	18	— 15

50 days for 65m development

Total days required = 81 days for 70m development

So days req. for 1m development = $\frac{81}{70} \approx 1.1$ days/m

Machineries/Equipments Required ———

- 1.} Cage or Gig Raising
- 2.} Alimak Raising
- 3.} Longhole Raising
- 4.} Raise Boring
- 5.} Hose Reel
- 6.} Conventional Raising

a) Why raising is preferred over winzing? Briefly discuss the different raising methods.

Winzing →

Winzing is the term for small-scale shaft sinking. Usually it is a form of internal mine development used, for example, to connect levels. Winzing has the following characteristics:

- Usually small diameter
- No room for mechanization, although in modern systems raise-boring machines can be used to develop winzes; however most are still sunk by hand.
- The possible range of sizes & depths are limited due to problems with muck removal.
- Muck removal is usually done by hand shovelling.
- Use of pilot holes makes winzing easier as deviation can be avoided if levels are connected by a pilot hole

- Ventilation is highly important, but usually poorly done
 - Secure ladders are essential for access
 - Small winches are used for raising and lowering small kipples
 - Typically a bench cut is used
 - Water can be a problem, but its impact can easily be overcome by using a pilot hole
 - All services need to be removed prior to a blast (ventilation ducting, water pumps and hoses, compressed air lines).
 - Hard physical work, especially in hot conditions
 - Vacuum-lift devices have been used successfully for muck removal.
- * Winzing is always a slower and more expensive undertaking than anticipated.

Raising →

A raise is a vertical or steeply inclined opening excavated from a lower level to a higher level (i.e., inverse winzing). The inclination is such that a ladder is req. for passage. Raises are used to gain access to ore, as travelling and supply routes, and as ventilation routes. Raises are also used as orepasses, wastepasses, or as slot raises to provide free faces for initial blasting of certain types of stopes.

Alimak Raising ———

Alimak raising is a development of cage raising using a rigid steel guide rail attached to one wall of the raise. No pilot hole

is req. & all the controls are mounted on the cage or at the base of the raise. The design of the system incorporates provision for air & water hoses, drilling equipment, blasting requirements, and accessories. Alimak raising has the capability of driving longer raises at faster rates than conventional methods.

The machine runs on a guide rail that incorporates a pin rack. The segmented guide rail is fastened to the side of the excavation with rock bolts. The Alimak methods has the following advantages over conventional systems :

- (i) It allows the driving of very long raises, vertical or inclined, straight or curved.
- (ii) When travelling to the head of the raise, personnel are well protected in a cage under the platform.
- (iii) The miners work from a platform that is easily adjusted for convenient height & angle.
- (iv) Risks involving gases are reduced, as the raise is ventilated after blasting with an air-water mixture supplied in the raise climber.
- (v) All equipment and material required can be taken to the head of the raise in the raise climber.
- (vi) Timbering is avoided, and if needed, rock bolts, with or without mesh, are used for support.
- (vii) The raise climber can be used for areas up to 8 m^2 .

b) Enumerate the different methods of direction and gradient control of a drift. Briefly discuss the working principle of automated guidance system for direction and gradient control used in TBM.

Direction Control Methods : —

- 1.> Manual Alignment : Using traditional surveying techniques with theodolites & lasers.
- 2.> Mechanical Systems : Employing mechanical guides along the drift path.
- 3.> Electronic Systems : Utilizing advanced electronic sensors and feedback mechanism.

Gradient Control Methods : —

- 1.> Level Instruments : Ensuring gradient accuracy with precise level instruments.
- 2.> Inclinometers : Continuous monitoring with inclinometers for any deviations.
- 3.> Automated System : Integrating automated systems for real-time adjustments.

Automated TBM Guidance system : —

- 1.> Laser-Based Tracking : A laser station determines the TBM's 3D position and orientation.
- 2.> Inclinometers : Dual-axis inclinometers measure the TBM's roll and tilt.
- 3.> Data Analysis : The system analyses data to guide the TBM along the designed alignment.

This guidance system ensures that the TBM maintains the correct direction and gradient as it excavates, providing real-time data to

the control cabin for necessary adjustments. The system is crucial for achieving the precision required in modern tunnel construction.

Construction Tolerance :-

It emphasizes the importance of constructing the tunnel within allowable construction tolerance, using various instruments like laser station, video Target, and dual axial inclinometers to measure the TBM's spatial position and orientation.

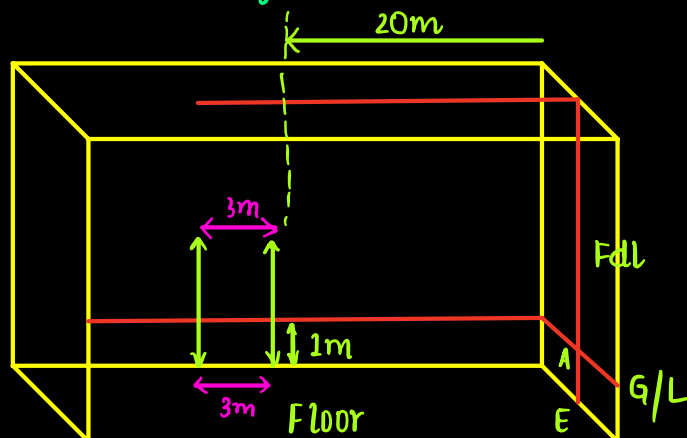
Post - Construction checks :-

After construction a "Wriggle survey" is conducted to check the as-built profile of the tunnel lining against construction and dimension tolerances, ensuring no infringement of the train's structure gauge.

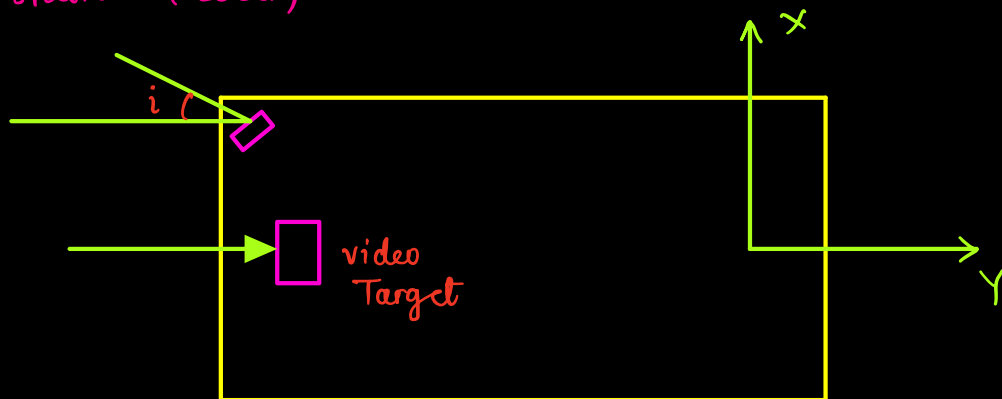
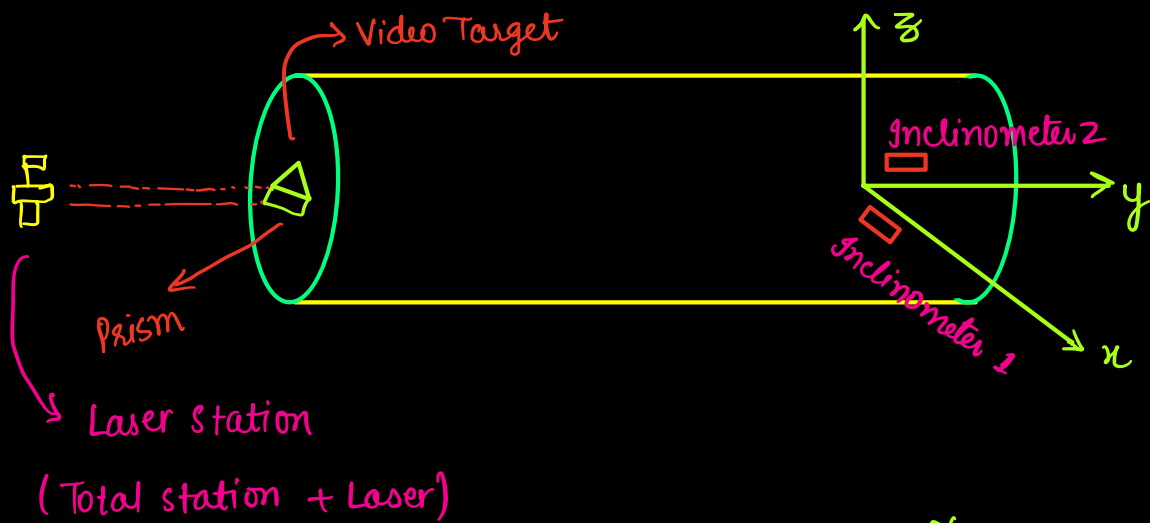
Maintenance of direction and gradient of a drift : —

① Peg System

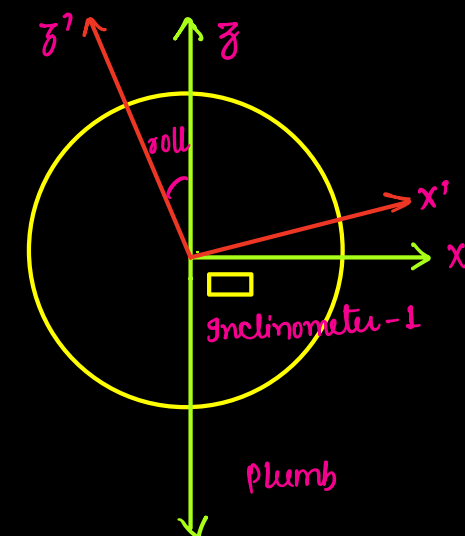
② Automated Guidance System



Peg System



$i=0 \rightarrow$ Means moving in y -direction
 $xy \rightarrow$ Plane (Plan view)



(Section view)

